

How the pipeline works

Selective betting with shrinkage

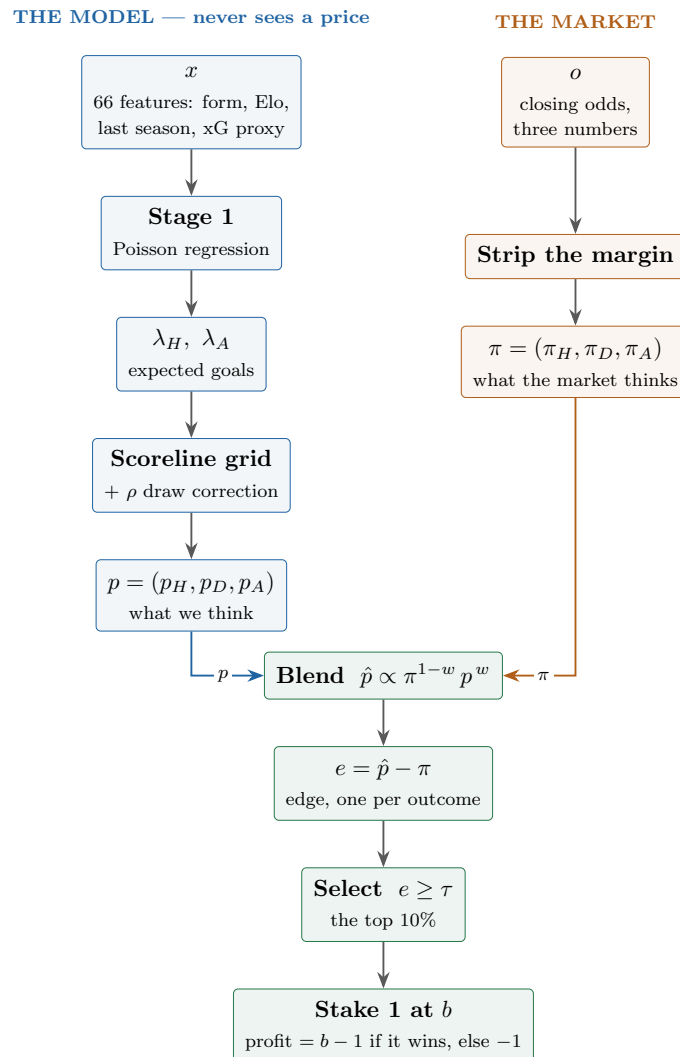
23 September 2026

The idea in one paragraph

A model picks the matches where it disagrees most with the bookmaker. Those are also the matches where the model is most likely to be wrong, because a large disagreement is usually a large error rather than a large insight. So the act of picking selects your own mistakes. The proposed fix is to pull the model's estimate part of the way toward the market price *before* picking anything. How far to pull is a single number, w , fitted from data.

The whole pipeline

Everything the model does is on the left, in blue. Everything the market contributes is on the right, in orange. They meet once, at the blend.



The blue column never sees a price. That is the whole reason the method can be tested: if stage 1 had the market among its inputs it would simply copy it, and pulling it toward the market

afterwards would correct nothing.

Every symbol

Symbol	What it is	Produced by	Learned?
x	66 market-blind features	the build / scripts	–
λ_H, λ_A	expected goals, each side	stage 1	yes
ρ	low-score draw correction	held-out fit, per league	yes
p	model probabilities	scoreline grid	no, arithmetic
o	closing average odds	the odds file	–
π	market probabilities	de-vigging o	no, arithmetic
w	shrinkage weight	fitted by log loss	yes
$\hat{p}$	corrected probabilities	the blend	no, arithmetic
e	edge, $\hat{p} - \pi$	subtraction	no
τ	selection threshold	90th percentile of e	from past seasons
b	best closing odds	the odds file	–

Only three things are learned: λ , ρ and w . Everything else is arithmetic or a look-up. The method has very few free parameters, deliberately.

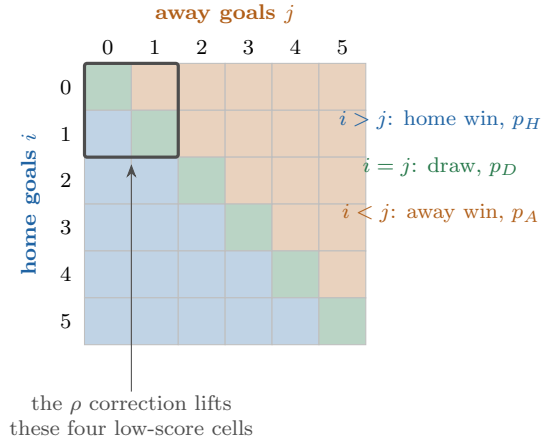
Stage 1: from features to a scoreline

Stage 1 does not predict who wins. It predicts how the match is played, as two expected-goal counts, then the outcome probabilities are computed from those.

Given λ_H and λ_A , the chance of the score being exactly i - j is a product of two Poissons, with a correction τ_ρ applied to the four low-scoring cells:

$$\Pr(i, j) = \tau_\rho(i, j) \cdot \frac{\lambda_H^i e^{-\lambda_H}}{i!} \cdot \frac{\lambda_A^j e^{-\lambda_A}}{j!}$$

Every cell with $i > j$ is a home win, $i = j$ a draw, $i < j$ an away win. Add the cells in each region and you have the three outcome probabilities.



Real football produces more 0-0 and 1-1 draws than two independent Poissons predict, because teams shut games down. That is what ρ repairs, and it is fitted separately for each league on held-out matches.

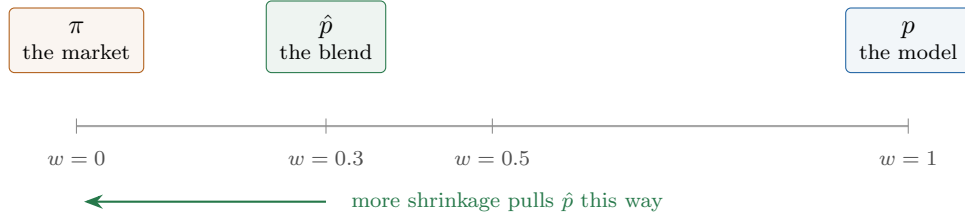
The market, and the blend

Odds are payout multipliers. Their reciprocals are implied probabilities, and they sum to more than one: that excess is the bookmaker's margin. Removing it gives π .

$$\underbrace{\frac{1}{2.10} + \frac{1}{3.50} + \frac{1}{3.80}}_{1.025} \longrightarrow \pi = (0.464, 0.279, 0.257)$$

The blend is a weighted geometric average, done in log-odds space so that it pulls hardest on the extreme probabilities — which is exactly where the model's error is largest:

$$\hat{p}_k \propto \pi_k^{1-w} p_k^w, \quad k \in \{H, D, A\}$$



w is fitted on out-of-fold predictions over every earlier season, by whichever value predicts held-out results best. We fitted it two further ways as a check: using only the matches we would actually bet, and with a different formula that shrinks the gap evenly rather than in log-odds. All three agreed.

One match, traced end to end

	Symbol	Value	What happened
1	x	66 numbers	Elo 1712 vs 1655, home side averaging 1.7 goals over five. No price anywhere in here.
2	λ_H, λ_A	1.62, 1.14	Stage 1's view of how the match will be played.
3	p	(0.49, 0.25, 0.26)	The scoreline grid summed into three regions.
4	o	(2.10, 3.50, 3.80)	Closing average odds; reciprocals sum to 1.025.
5	π	(0.464, 0.279, 0.257)	The same prices with the 2.5% margin removed.
6	$\hat{p}$	(0.472, 0.270, 0.258)	Blended at $w = 0.3$. The model moved the home probability by less than a point.
7	e	(+0.008, -0.009, +0.001)	The edge on each outcome.
8	τ	0.021	Read off earlier seasons. The best edge here is 0.008, well under it.
9	—	no bet	The normal outcome. We act on about one match in seven and abstain on the rest.

What actually happened

In every one of the five test seasons, the fitted weight came out

$$w = 0.$$

Put that into the blend and it collapses: $\hat{p} = \pi$, every edge in step 7 is exactly zero, and the corrected model *is* the market price.

That is not a bug and it is not a broken optimiser — we swept w by hand and every step away from zero makes the predictions worse. It is a finding. Stage 1 scores 1.0145 in log loss against the market's 0.9950, where knowing nothing scores 1.0770. The model closes about three quarters of the distance from ignorance to the closing price, and still adds nothing the price does not already contain.

The mechanism itself works. With no shrinkage the model expects a 127% return on the bets it selects and earns 2%; with full shrinkage it expects 10.8% and earns 6.2%. That collapse, from a 125-point gap to a 5-point one, is the winner's curse being removed exactly as the theory predicts. What the data denies us is anything to correct.